

AKSHATA TIWARI

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EDUCATION

Massachusetts Institute of Technology (MIT)

May 2027

Bachelor of Science in Computer Science/AI; Finance

Cambridge, MA

Relevant Coursework: Computer Vision, Deep Learning, Multimodal Modeling, Machine Learning, Statistics, Natural Language Processing, DS/Algorithms, Operating Systems, Software Construction (currently an LA)

Activities: HackMIT, AI@MIT, MIT AI Alignment, ML Performance Reading Group, Society of Women Engineers

SKILLS

Python, C/C++, Pytorch, Tensorflow, Numpy, Scikit-learn, Pandas, ONNX, JAX, CUDA, Jenkins, AWS, Linux

RELEVANT EXPERIENCE

MIT 3D Scene Representation Group

Jan 2026 - Present

Undergraduate Researcher, advised by Prof. Vincent Sitzmann

Cambridge, MA

- Studying 3D consistency in video world models by characterizing long-horizon drift between commanded camera motion and generated observations. Investigating model degradation of stable scene understanding over time.
- Developing an aggregate action-conditioned video-world model architecture and evaluation pipeline to reduce drift in 3D generation, measured through perceptual quality and estimated camera-pose error on RealEstate10K.

MIT Laboratory for Financial Engineering

Jun 2024 - Present

Undergraduate Researcher

Cambridge, MA

- Developing framework for vision-language models for visual pattern discovery in financial time series. Generated chart descriptions, clustering latents, explored VLM internals, and exploring iterative post-training strategies to improve recognition and pattern generalization; accepted at *ICLR Workshop on Advances in Financial AI: Agentic Systems*.
- Developing evaluation pipelines for discovering pattern families and exploring post-training on model generalization.

Citadel Securities

Jun 2025 - Aug 2025

Software Engineering Intern

New York City, NY

- Designed end-to-end execution engine to perform FINRA/SEC regulatory checks on front office trade data. Engineered pipelines with Apache Kafka, BigQuery, Celery-based multiprocessing for distributed, low-latency processing.
- Led system design, CI-tested unit coverage, deployed services to production via Jenkins and internal tooling. System is used by compliance and trading teams to reduce reporting errors and daily front-office regulatory overhead by 48%.

Nasdaq

Jan 2025

Machine Learning Research Intern

Boston, MA

- Trained hybrid model combining regression and random-forest classifier to predict OTC settlement failures with 98% acc. Used stratified k-fold cross-validation and GridSearchCV for hyperparameter optimization, applied feature engineering.
- Addressed data imbalances through SMOTE oversampling and presented end-to-end pipeline to business stakeholders.

SRI International

Jun 2024 - Dec. 2024

Machine Learning Research Intern

Menlo Park, CA

- Engineered and optimized diffusion models using PyTorch to handle out-of-distribution medical image inputs; developed and deployed distributed model training pipelines, generating over 50,000 high-quality medical images.
- Trained natural language models for medical data preprocessing, integrated diffusion models and transformer architectures to create data augmentation framework; shipped modular, tested code usable across modeling systems.

HONORS

Top 50 IEEE International Research Grant Winner

Honorable Mention: CA Science and Engineering Fair

USA Regeneron Science Talent Search Top 300 Scholar

USA Computing Olympiad Silver Division

1st Place Research: Orange County Science Fair

USA Biology Olympiad Honorable Mention

PUBLICATIONS

- **A. Tiwari**, A. Lo (2026) A VLM-Based Framework For Technical Analysis, *ICLR*, Adv. in Financial AI: Agentic Systems
- **A. Tiwari**, D. Haehn (2024). Stress Monitoring for Semi-Autonomous Systems, *NeurIPS 2024*, WiML Workshop
- **A. Tiwari**, B. Matejek, and D. Haehn (2024). Non-Invasive Stress Monitoring from Video. *IEEE International Symposium for Biomedical Imaging*, Vol. 21
- G. Su, **A. Tiwari**, T.K. Satish Kumar (2024). New Approaches for Winner Determination via MWVC, *AITA 2024*
- **A. Tiwari**. (2022), Effectiveness of Deep Learning for Erasing Satellite Streaks in Astronomical Photos, 2022 *IEEE Fourth International Conference on Transdisciplinary AI (TransAI)*, pp. 30-31
- **A. Tiwari**. (2022) Non-Invasive Stress Monitoring, poster presented at *Intl. Conf on Machine Learning (ICML)*